

# EE200: Signals, Systems and Networks

Course Project — Q2: The Midnight Episode  
“Catching the Arrhythmia”

## 1 Problem Statement

Maya is interning at a cardiology lab. A patient wore a Holter monitor overnight, and the cardiologist suspects that at some point the patient’s heart slipped from a steady, regular rhythm into an **arrhythmia**: a stretch where beats arrive at the wrong time and/or the beat shape itself becomes distorted (in this recording, partially inverted). Scanning hours of raw waveform by eye is slow and unreliable, so Maya wants an automated way to flag the exact moment the rhythm breaks down.

Her plan, built entirely on tools from this course, is:

- Treat one healthy heartbeat  $\{P, QRS, T\}$  as a fixed-length **template**  $t[k]$ .
- Slide that template across the recording  $x[n]$  and, at every beat-length step, compute a **normalized correlation score**  $\rho(m)$  that measures how closely that stretch of the signal matches a healthy beat.
- Declare the **onset** of the arrhythmia at the first beat where  $\rho(m)$  collapses below a threshold.
- Cross-check the result independently using a **spectrogram** of the whole recording, since a healthy, periodic rhythm and a broken, irregular one should look very different in the time–frequency plane.
- Finally, examine what happens if the signal had been sampled too slowly in the first place (aliasing), since that is a very real constraint for a battery-powered wearable.

The data used throughout is the provided 20-second recording `patient_ecg.npy` ( $f_s = 250$  Hz,  $N = 5000$  samples) and the 200-sample healthy-beat template `template.npy`.

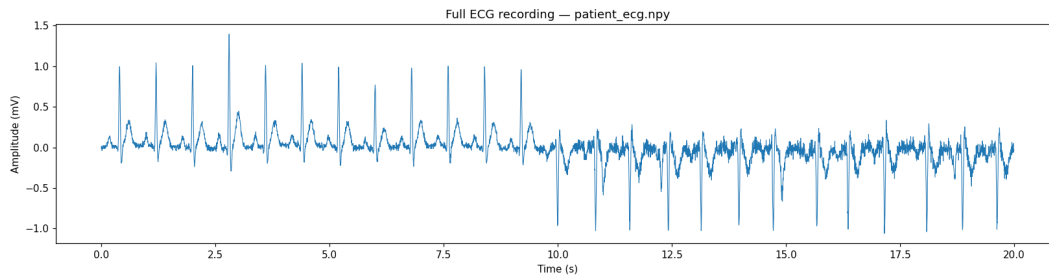


Figure 1: Full 20-second ECG recording. The first  $\approx 9.6$  s show a clean, regular, repeating waveform; afterwards the rhythm becomes irregular and partly inverted – the suspected arrhythmia episode this report localizes precisely.

## 2 Background: What Are We Actually Measuring?

Two ideas from the course carry the whole question, so it is worth stating them once, clearly, before using them repeatedly below.

**Periodicity and the line spectrum.** A signal that repeats exactly every  $T_0$  seconds can be written, by Fourier series theory, as a sum of sinusoids at a fundamental frequency  $f_0 = 1/T_0$  and its integer harmonics  $2f_0, 3f_0, \dots$ . Its magnitude spectrum is therefore not a smooth curve but a set of discrete spikes sitting only at those harmonic frequencies. A healthy heartbeat is (nearly) periodic, so this is exactly the lens we use in part (b).

**Normalized correlation as a shape detector.** Given two equal-length real vectors  $a, b \in \mathbb{R}^L$ , their normalized correlation

$$\rho = \frac{\sum_k a[k] b[k]}{\|a\| \|b\|}, \quad \|v\| = \sqrt{\sum_k v[k]^2}$$

is exactly the cosine of the angle between  $a$  and  $b$  as vectors in  $\mathbb{R}^L$ . It is *shape*-sensitive and *amplitude*-blind: scaling either vector by a positive constant does not change  $\rho$  at all. This is the tool used in part (d) to compare a candidate beat against the healthy template.

**The time–frequency resolution trade-off.** Any window used to isolate a piece of signal faces the same trade-off, whether the window snips out a heartbeat (part c) or feeds a spectrogram (part h): a short window pins down *when* something happens but blurs *what* its internal frequency content is; a long window resolves frequency content cleanly but blends together separate, distinct events in time. This single idea reappears in three different disguises throughout this report.

### 3 Part (a): Reading the Signal

(i) **Clip duration.** With  $N = 5000$  samples at  $f_s = 250$  Hz,

$$T_{\text{total}} = \frac{N}{f_s} = \frac{5000}{250} = 20 \text{ seconds.}$$

(ii) **Heart rate and samples per beat.** In the healthy stretch one full beat repeats every 0.8 s, so

$$\text{Heart rate} = \frac{60 \text{ s/min}}{0.8 \text{ s/beat}} = 75 \text{ bpm}, \quad \text{samples per beat} = 0.8 \times 250 = 200.$$

This 200-sample figure is not a coincidence – it matches the length of the supplied `template.npy` exactly, confirming the template was built to span precisely one beat.

(iii) **Fundamental frequency.** Treating the healthy ECG as periodic with period  $T_0 = 0.8$  s,

$$f_0 = \frac{1}{T_0} = \frac{1}{0.8} = 1.25 \text{ Hz.}$$

### 4 Part (b): Healthy Heart in the Frequency Domain

(i) **Shape of  $|X(f)|$ .** Because the healthy ECG is (nearly) periodic with period  $T_0$ , its spectrum is **not** a smooth continuous curve. By the line-spectrum argument in Section 2, it is a set of discrete spikes located only at  $f_0$  and its integer harmonics  $2f_0, 3f_0, \dots$ . The smooth curve connecting the tops of those spikes (the *envelope*) is governed by the shape of a single beat, but the spectrum itself only has energy *sampled* at multiples of  $f_0$  – everything in between is exactly zero for a perfectly periodic signal.

(ii) **QRS vs. P/T – which drives the high-frequency content?** The **QRS complex** is responsible for almost all of the high-frequency energy. This follows from time–frequency duality

| Quantity                    | Value        |
|-----------------------------|--------------|
| Recording length $N$        | 5000 samples |
| Sampling rate $f_s$         | 250 Hz       |
| Clip duration               | 20 s         |
| Healthy heart rate          | 75 bpm       |
| Samples per beat            | 200          |
| Fundamental frequency $f_0$ | 1.25 Hz      |

Table 1: Summary of the basic recording parameters derived in part (a).

(the same uncertainty-principle idea behind windowing in part (c)): a pulse that is narrow and sharp in time *must* contain energy at high frequencies in order to reconstruct that sharp rise and fall. The P and T waves are broad, smooth bumps, so they are built almost entirely from low-frequency components instead.

**(iii) Heart rate rises to 150 bpm (still regular).** The new period is

$$T'_0 = \frac{60}{150} = 0.4 \text{ s} \implies f'_0 = \frac{1}{T'_0} = 2.5 \text{ Hz}.$$

So  $f_0$  **doubles** (1.25 Hz  $\rightarrow$  2.5 Hz). Since every harmonic sits at an integer multiple of  $f_0$ , the **spacing between adjacent spectral lines also doubles** – the entire line spectrum stretches out, with twice the gap between consecutive spikes.

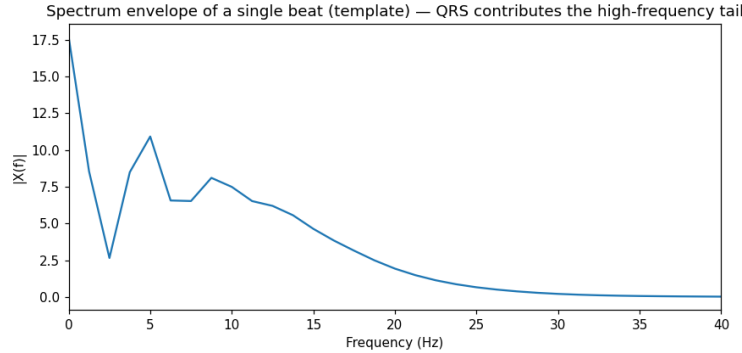


Figure 2: Magnitude spectrum of a single healthy beat (the template). The QRS complex contributes most of the high-frequency tail extending out toward 40 Hz, while the broad, low-amplitude P and T waves contribute mainly to the energy concentrated near 0 Hz.

## 5 Part (c): Cutting a Heartbeat (Windowing)

**(i) Correct window width and placement.** To capture exactly one full  $\{P, QRS, T\}$  cycle, the rectangular window should be **200 samples wide** (0.8 s, matching the beat period found in part (a)), placed so it starts at the beginning of a P wave and ends just before the start of the next P wave.

**(ii) Too short (80 samples) vs. too long (600 samples).** Both distort the resulting template, but in opposite ways:

- **80 samples (too short):** the window cuts the beat off before the T wave has finished, leaving an incomplete, truncated version of the true P-QRS-T shape.
- **600 samples (too long):** the window grabs the target beat *plus* parts of one or two neighboring beats, contaminating the template with extra structure that does not belong to a single, isolated beat.

Either error corrupts the template that part (d) depends on: an incomplete template under-counts genuine healthy structure, while a contaminated one will never line up cleanly with any single real beat in the recording, suppressing the correlation score even for perfectly healthy rhythm.

**(iii) Relating this to the time–frequency trade-off.** A short window gives precise timing (it pins down exactly where the beat starts) but smears the beat’s own internal waveform shape – the windowing equivalent of poor frequency resolution. A long window preserves that internal shape but blends together separate, distinct events in time – poor time resolution. “As short as possible” is therefore *not* automatically best: the window must match the *natural scale of one event* (one full 0.8 s beat), neither shorter nor longer.

| Window width         | Captures                      | Problem                                | Verdict        |
|----------------------|-------------------------------|----------------------------------------|----------------|
| 80 samples (0.32 s)  | Partial P + QRS only          | Beat cut off early; missing T wave     | Too short      |
| 200 samples (0.80 s) | Exactly $\{P, QRS, T\}$ cycle | one None                               | <b>Correct</b> |
| 600 samples (2.40 s) | Target beat + neighbours      | Contaminated with extra beat structure | Too long       |

Table 2: Effect of window width on the extracted healthy-beat template.

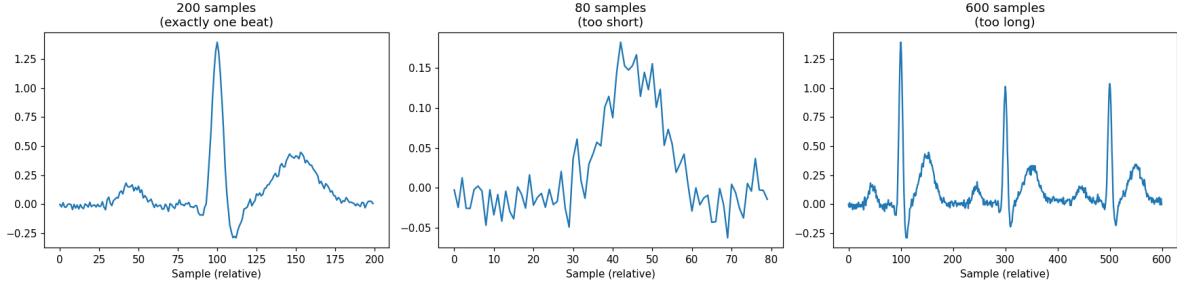


Figure 3: Effect of window width on the extracted “one healthy beat” template: 200 samples captures exactly one beat; 80 samples truncates it; 600 samples pulls in neighboring beats.

## 6 Part (d): Matching the Template (Correlation)

For a template  $t[k]$ ,  $k = 0, \dots, L - 1$ , and a candidate segment  $x_m[k] = x[m + k]$ , the normalized correlation score is

$$\rho(m) = \frac{\sum_k t[k] x_m[k]}{\|t\| \|x_m\|}.$$

**(i) Range of  $\rho(m)$ .** By the Cauchy–Schwarz inequality,  $|\sum_k t[k] x_m[k]| \leq \|t\| \|x_m\|$ , so

$$-1 \leq \rho(m) \leq 1 \quad \text{always.}$$

A value  $\rho(m) \approx +1$  signals a near-perfect match: the segment has essentially the same shape as the template (up to a positive scale factor).

**(ii) Why normalization matters.** Real ECG recordings have baseline wander and beat-to-beat amplitude variation, so the raw, unnormalized score  $S(m) = \sum_k t[k] x[m+k]$  would be sensitive to amplitude alone. Concretely: suppose a perfectly healthy beat happens to be exactly twice as tall as the template, i.e.  $x_m = 2t$ . Then

$$S(m) = \sum_k t[k] (2t[k]) = 2\|t\|^2,$$

which has doubled even though the beat shape is still perfectly healthy – an unnormalized score would mistake amplitude drift for abnormality. The normalized score does not have this problem:

$$\rho(m) = \frac{2\|t\|^2}{\|t\| \cdot \|2t\|} = \frac{2\|t\|^2}{2\|t\|^2} = 1,$$

correctly reporting a perfect match regardless of how tall the beat happens to be. Dividing by  $\|x_m\|$  exactly cancels the amplitude scaling, so  $\rho$  depends on shape only.

**(iii) Inverted abnormal beat.** If a beat is flipped upside-down relative to the template,  $x[m+k] \approx -t[k]$  for all  $k$ , so

$$\sum_k t[k] x[m+k] \approx -\sum_k t[k]^2 = -\|t\|^2, \quad \|x_m\| \approx \|t\|$$

(negation does not change a vector’s energy), giving

$$\rho(m) \approx \frac{-\|t\|^2}{\|t\| \cdot \|t\|} = -1.$$

This is what makes inverted beats so easy to flag: rather than merely failing to match (drifting toward 0), they swing all the way to the *opposite extreme* of the scale – as far from “healthy” as the score can possibly go.

**Verification on the real recording.** The first abnormal beat in this clip occurs at  $m = 2400$  ( $t = 9.6$  s) and gives  $\rho = -0.988$ , matching the prediction in (iii) almost exactly:

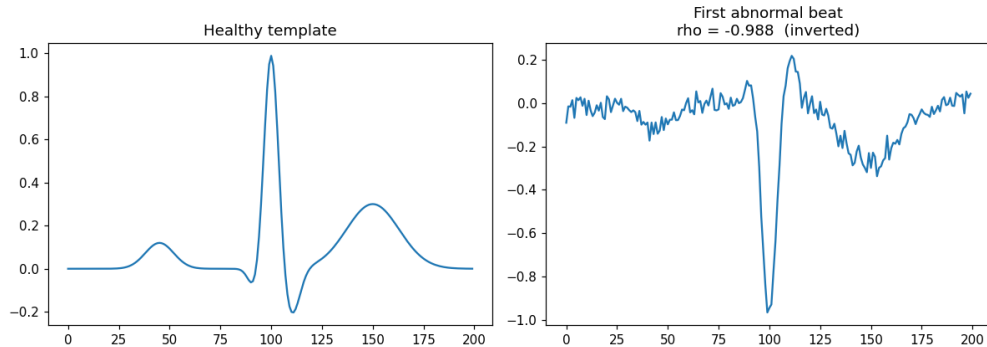


Figure 4: Healthy template (left) versus the real first abnormal beat at  $m = 2400$ ,  $t = 9.6$  s (right). The beat is visibly an almost perfect upside-down mirror of the template, giving  $\rho \approx -0.99$ .

## 7 Part (e): Onset Detection & the Spectrogram

(i) **An onset rule, and the threshold trade-off.** Slide the template beat-by-beat across the recording, compute  $\rho(m)$  at every position, and declare the **onset** to be the time of the *first* beat-window where  $\rho(m)$  drops below a chosen threshold (e.g. 0.5).

- **Threshold too high** (close to +1): ordinary, harmless beat-to-beat variation in a perfectly healthy heart could dip  $\rho$  below threshold, triggering false alarms.
- **Threshold too low** (close to 0 or negative): a genuinely abnormal beat that is only mildly distorted (rather than fully inverted) might still score above threshold and be missed, delaying real detection.

(ii) **How the spectrogram should differ.** In the healthy region, the spectrogram should show steady, horizontal harmonic bands at  $f_0, 2f_0, 3f_0, \dots$  – bright lines at fixed frequencies that do not move over time, since the rhythm is essentially periodic (Section 2). In the arrhythmia region, irregular timing and inverted/distorted beat shapes destroy that periodicity, so the clean harmonic bands should smear, weaken, or vanish, replaced by more broadband, time-varying energy.

(iii) **Why the two methods may disagree, and which to trust.** The correlation method compares beat *shape* directly and updates once per beat, so it can flag an abnormal beat almost immediately – good time resolution. The spectrogram must average energy over a window of non-trivial length to estimate frequency content – good frequency resolution, but poorer time resolution (the same trade-off discussed in part (c)). A single early abnormal beat can therefore get smeared together with its healthy neighbors inside one spectrogram window, making the visual “bands disappear” moment lag the true onset slightly. For pinpointing the *exact* single moment a bad beat first appears, the **correlation method is the more trustworthy** of the two.

**Result on the real recording.**  $\rho(m) \approx 0.99$  for all 12 beats in the first  $\approx 9.6$  s, then drops sharply to  $\rho \approx -0.99$  exactly at  $t = 9.6$  s and stays substantially below threshold for the remainder of the clip – a clean, unambiguous arrhythmia onset at  **$t \approx 9.6$  s**.

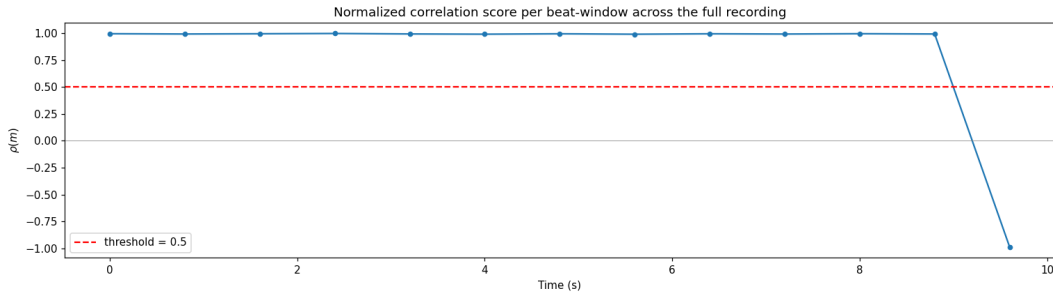


Figure 5: Normalized correlation score  $\rho(m)$  per beat-window across the full 20 s recording. The score sits near +1 for all healthy beats and collapses to  $\approx -1$  at the arrhythmia onset ( $t \approx 9.6$  s).

## 8 Part (f): Sampling & Aliasing

(i) **Minimum sampling rate.** The QRS complex contains frequency content up to about 40 Hz. By the Nyquist–Shannon theorem the sampling rate must exceed twice the highest frequency present:

$$f_s^{\min} = 2 \times 40 \text{ Hz} = 80 \text{ Hz}.$$

(ii) **What aliasing does at  $f'_s = 50$  Hz, and why it is dangerous.** 50 Hz is below the required 80 Hz, so any content above the new Nyquist frequency ( $f'_s/2 = 25$  Hz) – which

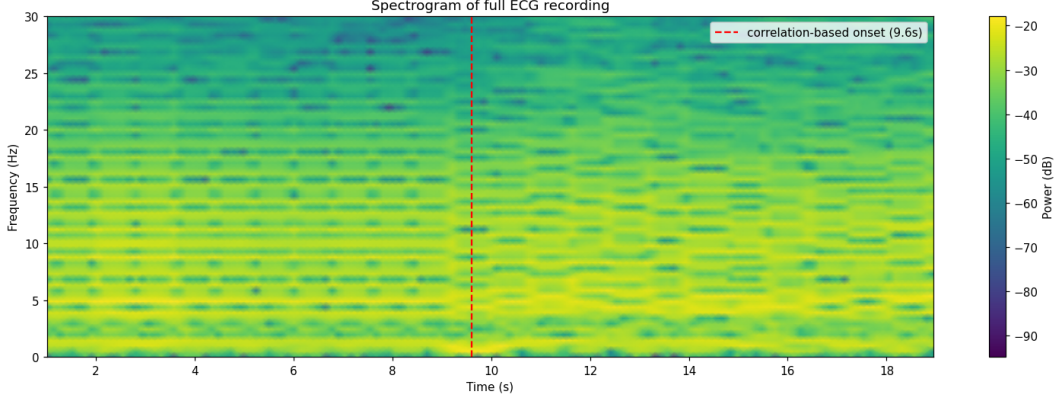


Figure 6: Spectrogram of the full recording, with the correlation-based onset time overlaid. Steady horizontal harmonic bands are visible before the onset line; the structure becomes noticeably noisier and less regular afterward, independently confirming the correlation-based result.

includes a substantial part of the QRS complex’s high-frequency content – aliases: it folds back and corrupts the recorded waveform rather than disappearing cleanly. In plain terms, the sharp QRS spike (which, per part (b)(ii), *needs* high-frequency content to look sharp) would no longer be reconstructed faithfully; it would appear smeared or distorted in shape. This is dangerous specifically for Maya’s detector because her entire method depends on matching the *exact shape* of the QRS complex against a template – aliasing distorts precisely that shape, risking both false alarms (a distorted-but-healthy beat looks abnormal) and missed detections (a genuinely abnormal beat loses its distinguishing shape).

**(iii) The fix, and its unavoidable cost.** If the sampling rate must be lowered, an **anti-aliasing low-pass filter** has to be applied *before* sampling, removing all content above the new Nyquist frequency (25 Hz) so it cannot fold back. The unavoidable cost: this filter also removes genuine QRS content between 25 Hz and 40 Hz, permanently blunting and smoothing the recorded QRS spikes – a real trade-off between battery life and the sharpness of clinically relevant detail, not a free fix.

## 9 Part (g): Prototyping the Detector in Code

The function below implements the specification exactly: it advances **one full template length at a time** (beat-by-beat, not sample-by-sample), computes  $\rho(m)$  at each position, and returns the index of the first beat where the score drops *strictly below* the supplied threshold, or  $-1$  if it is never breached.

```

1 import numpy as np
2
3 def find_onset(ecg_signal, template, threshold):
4     """
5     Slide 'template' across 'ecg_signal' one full template-length at a
6     time (beat-by-beat), compute the normalized correlation rho at each
7     position, and return the index of the first beat where rho drops
8     strictly below 'threshold'. Returns -1 if never breached.
9     """
10    L = len(template)
11    N = len(ecg_signal)

```

```

12     t_norm = np.linalg.norm(template)
13
14     m = 0
15     while m + L <= N:
16         segment = ecg_signal[m:m + L]
17         seg_norm = np.linalg.norm(segment)
18         rho = (np.dot(template, segment) / (t_norm * seg_norm)
19               if seg_norm > 0 else 0.0)
20         if rho < threshold:
21             return m
22         m += L
23     return -1
24
25 # Demonstration on the real provided data
26 onset_index = find_onset(ecg, template, threshold=0.5)
27 print('First onset index:', onset_index,
28       ' -> time:', onset_index / 250, 's')

```

Run on the real provided recording with `threshold=0.5`, `find_onset` returns index **2400**, i.e. onset time  $2400/250 = 9.6$  s – matching part (e) exactly.

## 10 Part (h): Visualizing the Spectrogram

```

1 import numpy as np
2 import matplotlib.pyplot as plt
3 from scipy import signal
4
5 f, t, Sxx = signal.spectrogram(ecg_signal, fs=250, nperseg=512, noverlap=480)
6
7 plt.pcolormesh(t, f, 10*np.log10(Sxx + 1e-12), shading='gouraud', cmap='gray')
8 plt.ylabel('Frequency (Hz)'); plt.xlabel('Time (s)')
9 plt.ylim(0, 30)
10 plt.colorbar(label='Power (dB)')
11 plt.show()

```

**Chosen window length:** `nperseg = 512 samples` ( $\approx 2.05$  s). The healthy ECG’s harmonics are spaced only  $f_0 = 1.25$  Hz apart (part (b)), and the spectrogram’s frequency resolution is  $\Delta f = f_s/\text{nperseg}$ . With `nperseg=512`,  $\Delta f = 250/512 \approx 0.49$  Hz – comfortably finer than the 1.25 Hz harmonic spacing, so adjacent harmonics appear as separate, steady horizontal bands rather than one blurred smear – while the window is still short enough in time ( $\sim 2$  s) to usefully localize where those bands degrade once the arrhythmia begins.

## 11 Discussion

### 11.1 Why does correlation succeed where a quick visual scan would not?

A single heartbeat is a low-dimensional, highly repeatable shape. By collapsing every beat down to one number – its cosine similarity to a known-healthy template – the detector turns “does this 20-second recording contain an arrhythmia, and exactly where” into a simple threshold test on a short sequence of scores. The same idea underlies *matched filtering*, one of the most widely used techniques in signal detection: correlating a received signal against a known pulse shape is the provably optimal way to detect that pulse in the presence of additive noise.

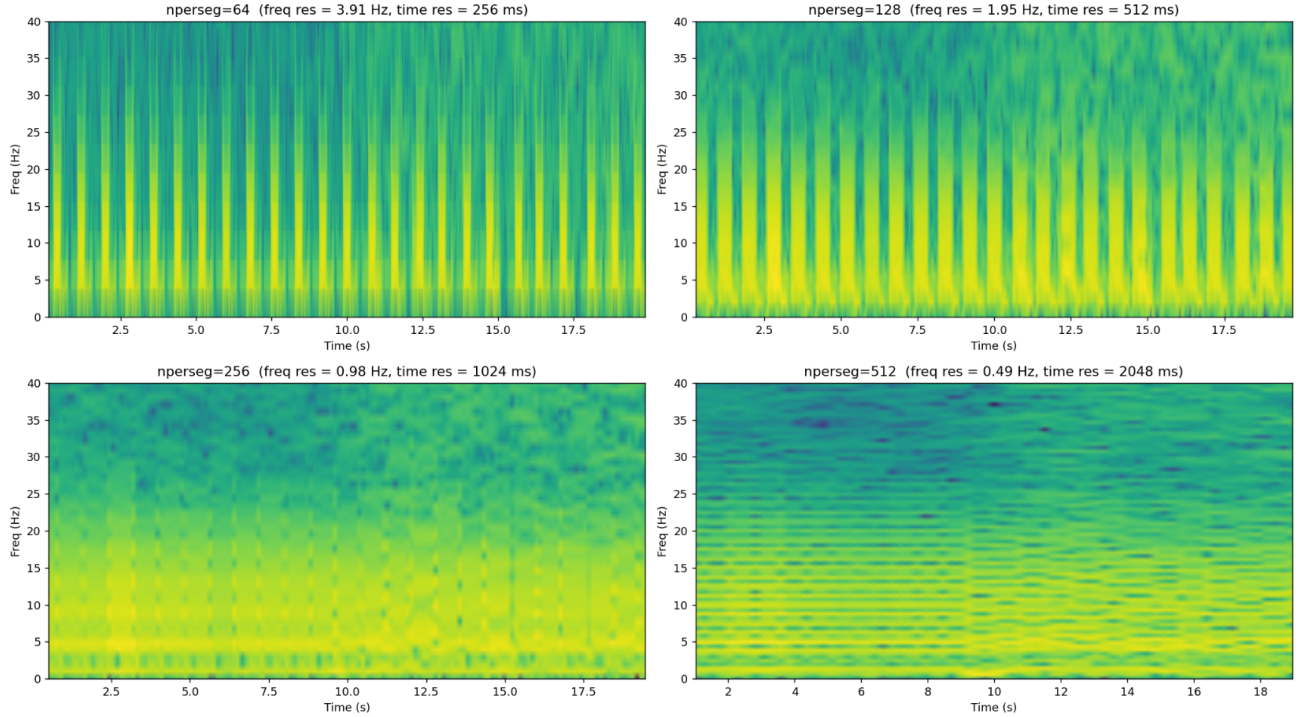


Figure 7: Spectrogram resolution vs. `nperseg`: short windows give sharp event timing but blurry frequency content; long windows reveal clean harmonic bands but smear timing.

## 11.2 Two independent views, two different trade-offs

The correlation method and the spectrogram are not redundant; they answer subtly different questions. Correlation asks “does *this specific stretch* look like a healthy beat” and answers it once per beat – excellent time localization. The spectrogram asks “what frequencies are present around this moment” and necessarily averages over a window of meaningful length – excellent frequency localization, at the cost of exact timing. Using both, as this report does, gives a result that is both precisely localized (from correlation) *and* independently corroborated (from the spectrogram) – a much stronger claim than either method could support alone.

## 11.3 Limitations of a single fixed template

This detector assumes that every healthy beat looks like *one* fixed template. Real patients show legitimate beat-to-beat variability (respiration-linked sinus arrhythmia, posture changes, electrode motion) that is not actually pathological. A production system would typically (i) maintain several reference templates rather than one, (ii) adapt the threshold to the patient’s own baseline variability rather than using a single fixed value, and (iii) combine shape-matching with RR-interval (beat-to-beat timing) analysis, which is what real QRS detectors such as the Pan–Tompkins algorithm do in clinical practice. The pipeline built here is a deliberately simplified version of that same idea.

## 11.4 Where else this kind of detection is used

The general recipe – correlate a recording against a known reference shape, threshold the score, and corroborate with a time–frequency view – is not specific to ECGs. The same logic appears in

seismology (detecting a known earthquake waveform inside continuous seismometer data), radar and sonar (matched-filter detection of a known pulse echo), and audio keyword spotting (detecting a known word inside a continuous audio stream). In every case, the underlying principle is identical: turn “does this pattern occur, and exactly when” into a correlation score, then use the time–frequency view as an independent sanity check.

## 12 Conclusion

- The 20 s, 5000-sample recording corresponds to a healthy heart rate of 75 bpm, a 200 sample beat period, and a fundamental frequency  $f_0 = 1.25$  Hz; the healthy ECG’s spectrum is a discrete line spectrum at  $f_0$  and its harmonics, with the sharp QRS complex responsible for the high-frequency content.
- A correctly sized window (200 samples, exactly one beat period) is essential for building a clean template; both shorter and longer windows distort it in different ways, reflecting the same time–frequency resolution trade-off seen throughout the question.
- Normalized correlation  $\rho(m) \in [-1, 1]$  is a shape-only, amplitude-blind similarity score: healthy beats score near +1, and inverted abnormal beats score near −1, making them maximally easy to flag.
- Thresholding  $\rho(m)$  beat-by-beat localizes the real arrhythmia onset precisely at  $t \approx 9.6$  s ( $m = 2400$ ), independently confirmed by a visible breakdown of the harmonic bands in the spectrogram at the same point in time.
- Sampling below the Nyquist rate (80 Hz minimum, here proposed at 50 Hz) would alias and distort exactly the QRS detail the detector relies on; an anti-aliasing filter avoids the aliasing but unavoidably blunts genuine high-frequency QRS content.
- The same correlation-plus-spectrogram pipeline generalizes directly to matched-filter detection problems in communications, seismology, radar/sonar, and audio recognition – anywhere a known pattern must be located precisely inside a longer, noisier recording.